

Diagnosis and Differential Diagnosis of AGEP

Skin biopsy including a pustule is useful for confirming the diagnosis of **acute generalized exanthematous pustulosis (AGEP)**. Histological findings typically include spongiform subcorneal and/or intraepithelial pustules, oedematous papillary dermis, and perivascular infiltrates composed of neutrophils and some eosinophils. In certain cases, necrotic keratinocytes and leucocytoclastic vasculitis may also be observed. The presence of psoriasiform acanthosis, papillomatosis, and tortuous or dilated vessels suggests the main differential diagnosis: **generalized pustular psoriasis (GPP)**.

A personal or family history of psoriasis is reported in 7%–10% of AGEP patients. The primary differential diagnosis remains GPP. Pustules in GPP are often larger, with a slower onset and longer duration compared to those in AGEP. Other rare pustular dermatoses include subcorneal pustulosis (Sneddon-Wilkinson disease), IgA pemphigus, pustular vasculitis, pustular Sweet syndrome, and staphylococcal scalded skin syndrome (SSSS). In GPP and other non-drug-induced pustular conditions, there is typically no history of recent drug exposure or adverse cutaneous drug reaction.

AGEP can also be confused with other drug-induced skin reactions. Pustules may appear in **DRESS syndrome**, particularly on the face and occasionally in large body folds. Similarly, pustules can occur in other drug-related eruptions involving intertriginous areas, such as **symmetrical drug-related intertriginous and flexural exanthema (SDRIFE)** and **Baboon syndrome**.

Etiology of AGEP

AGEP is drug-induced in approximately **90% of cases**. The most common causative agents include:

- **Antibiotics:** pristinamycin, aminopenicillins, quinolones, sulfonamides
- **Antimalarials:** (hydroxy)chloroquine

- **Antifungals:** terbinafine
- **Calcium channel blockers:** diltiazem

Notably, a French case series documented 19 instances of AGEF linked to **iodinated contrast media**. The time between drug initiation and onset of AGEF is typically short, usually within 1–5 days.